

Rivet — Poster

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Verified multimodal ad creation on one AMD Radeon GPU

AMD AI DevMaster Hackathon 2026 · Track 1 · Radeon PRO W7900 · ROCm 7.2

The problem

A brand can generate a thousand advertisements in an afternoon and ship none of them.

Not because the quality is poor — because a generative model redraws whatever it is given. It bends the logo, invents a button the product does not have, and renders text as plausible glyphs that spell nothing. Marketing cannot show that to customers. Legal cannot approve it. So the output goes back to a designer to rebuild by hand, and the generative step has saved nobody any time.

The usual answer is to ask for trust: better prompts, better models, a human at the end. Review does not scale to a thousand assets, and trust is not evidence.

What Rivet does

A product photo, a brand kit and a short brief become a **verified three-scene vertical advertisement** — on one Radeon GPU, with no network access.

The model generates the background, and nothing else. The product cutout, the logo and every character of text are composited afterwards, deterministically, from the original files. This is structural rather than advisory: the compositor reads the source assets directly, and the audit proves what landed in the frame is pixel-identical to what came in.

Every export carries a Campaign Receipt — input hashes, seeds, per-stage wall time, peak VRAM, every check with its observed value, and any repair applied. A pack ships with a manifest hashing every file in it.

Ten checks on every scene

	verifies
A01	product and logo match the brand-confirmed assets, by sha256
A02	the logo in the frame is the real logo, pixel for pixel
A03	rendered copy equals approved copy
A04	the background stays on-brand
A05	safe-area geometry, and that no text overflowed
A06	the product occupies enough of the frame
A07	no forbidden claims, all required phrases — including narration
A09	the product in the frame is the real product, pixel for pixel
A10	text is legible against what is actually behind it
A08	<i>advisory</i> — semantic fit judged by Qwen3-VL, never blocks

If a deterministic check fails, **no pack is written**. The project moves to `needs_repair`. When a claims check fails, Rivet rewrites the offending copy, recomposites, re-audits and records what changed — and the audit runs *before* the video is rendered, so the film is built from the repaired frames rather than certified separately from them.

Measured on the Radeon PRO W7900

gfx1100 · 49136 MB · ROCm 7.2.53211 · PyTorch 2.9.1

	seconds	audit
Cold — plan, generate, audit, render, pack	67.8	27/27
Hot — models already resident	41.2	27/27
Offline — every outbound socket blocked	67	27/27

Peak VRAM 9168 MB of 49136. One heavy model is held at a time; 81% of the card stays free.

Model residency — the optimisation

SDXL rebuilt its pipeline and VAE from disk for every scene. Heavy models now pass through a residency scheduler: acquiring a different model releases the previous one first, so exactly one is resident and the VRAM headroom rule holds by construction.

arm	median total
resident	42.2s
reload per stage	55.8s

13.7s faster — 24% end to end, 32% on generation. Arms alternate after a discarded warmup: an unalternated pair hands the second arm a warm page cache worth ~17s, more than the effect itself. The first attempt reported residency as *slower*; it was measuring run order and was discarded.

Determinism

One content digest — 347606b58a93ba16 — across **all eleven runs**, including both residency arms. It covers rendered pixels, seeds and audit observations, so only identical output satisfies it.

Reproduce it

```
uv sync --extra gpu
make offline-demo      # passing export, network blocked
make test-golden       # byte-identical compositing
uv run rivet benchmark --fixture fixtures/kora-arc --mode residency
```

Every number here is produced by these commands. Nothing is quoted from a private session.

Qwen3-VL-4B plans · **SDXL** backgrounds · **SAM 2.1** cutout · **Kokoro-82M** narration · **Whisper** brief · deterministic Python composites, audits and signs.